BRIAN NGUYEN
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SKILLS

Programming Languages: C++, Python, CUDA, Java, Bash
Perception: Computer Vision, OpenCV, Object Detection, Multi-Object Tracking, Kalman Filtering, Optical Flow
ML and Edge Deployment: PyTorch, TensorRT, CUDA Graphs, ONNX Runtime, HailoRT, LiteRT, INT8/FP16 Quantization
Hardware: NVIDIA Jetson Orin, Hailo-8/10H NPUs, Qualcomm Boards, Raspberry Pi 5
Systems and Tooling: Bazel, GStreamer, Docker, GitLab CI, GoogleTest, Multithreading, Linux, Git

EXPERIENCE

Modern Intelligence - Austin, TX
Junior Software Engineer Nov 2025 - Present
Software Engineering Intern Jun - Oct 2025

- Built a native TensorRT backend for a real-time C++ inference engine that cut latency **2-3x** across the Jetson Orin family and used CUDA Graphs with persistent GPU buffers to reduce p50 latency another **21%**
- Rewrote image pre/post-processing for 5 detector families in native C++ to speed up preprocessing **6.8x** on Raspberry Pi 5 and **2x** end-to-end throughput on Hailo-8
- Accelerated PyTorch training of transformer-based object detectors **2.4-3.5x** per step on multi-GPU clusters by rebuilding the GPU data pipeline and removing CPU-GPU syncs to enable full-graph torch.compile
- Redesigned the loss and evaluation pipeline to halve peak GPU memory and eliminate out-of-memory crashes while cutting per-epoch evaluation time by over **90%** with no change in mAP
- Ported the perception stack to **4** new edge targets spanning Hailo-8/10H NPUs, VOXL 2, and Raspberry Pi 5 using hermetic Bazel cross-compilation and hardware-in-the-loop CI
- Replaced FFmpeg with low-latency GStreamer video pipelines for RTSP/RTP/SRT/V4L2 sources with Jetson hardware decoding and automatic stream recovery
- Improved multi-target tracking with **~3x** faster GPU optical flow and quicker detection-to-track matching plus per-track class voting to reduce label flicker
- Implemented target geolocation that converts pixel detections into GPS coordinates from MISB KLV drone telemetry and built an end-to-end Cursor-on-Target accuracy test

College Readiness Mentor - UT Austin May - Aug 2024, May - Aug 2025

- Guided 250+ incoming students during their transition to college through weekly office hours and online support
- Ran tutoring and Q&A sessions for students in the College of Natural Sciences

Harpak Lab - *Undergraduate Student Researcher* - UT Austin Jan - May 2024

- Conducted genome-wide analyses using Bash, R, and PLINK on the TACC supercomputer system
- Participated in weekly lab meetings to discuss papers on polygenic scores for insights into current advancements

PROJECTS

Visionnaire | Python, PyTorch, Triton, Qdrant, FastAPI, React

- Served one CoCa ViT-L-14 model on Triton for both image captioning and 768-dim Qdrant search embeddings
- Deployed an ARM64 cloud backend calling home-GPU inference via Cloudflare Tunnel with auto-rollback CI/CD

Fine-Grained GPU Synchronization | CUDA & C++

- Reimplemented ASPLOS '19 fine-grained GPU locking with shared-memory locks and a lock-free MPMC queue
- Achieved up to **6.22x** speedup over sequential on a GPU hash table, **2.2x** faster than a global-lock baseline

EDUCATION

The University of Texas at Austin Aug 2021 - May 2025
BSA in Computer Science - GPA: 3.72 | *Certificate in Applied Statistical Modeling*
Relevant Courses: Algorithms, Computer Architecture, Operating Systems, Compilers, Concurrency, Machine Learning